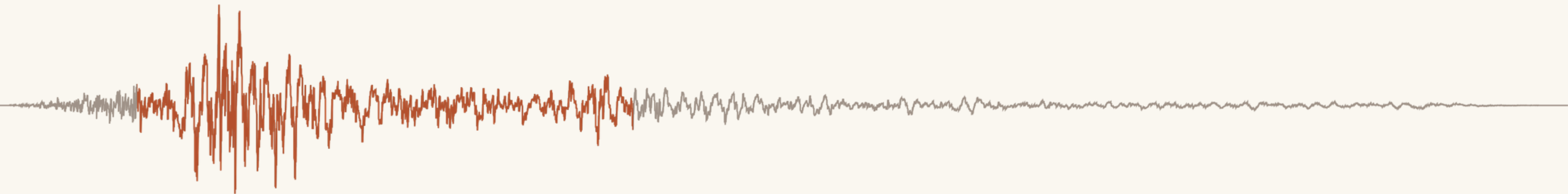


Lucky Seismic Strike



TEAM NP.INTERP(1, -1)
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INTRODUCTION

What we built

Lucky Seismic Strike

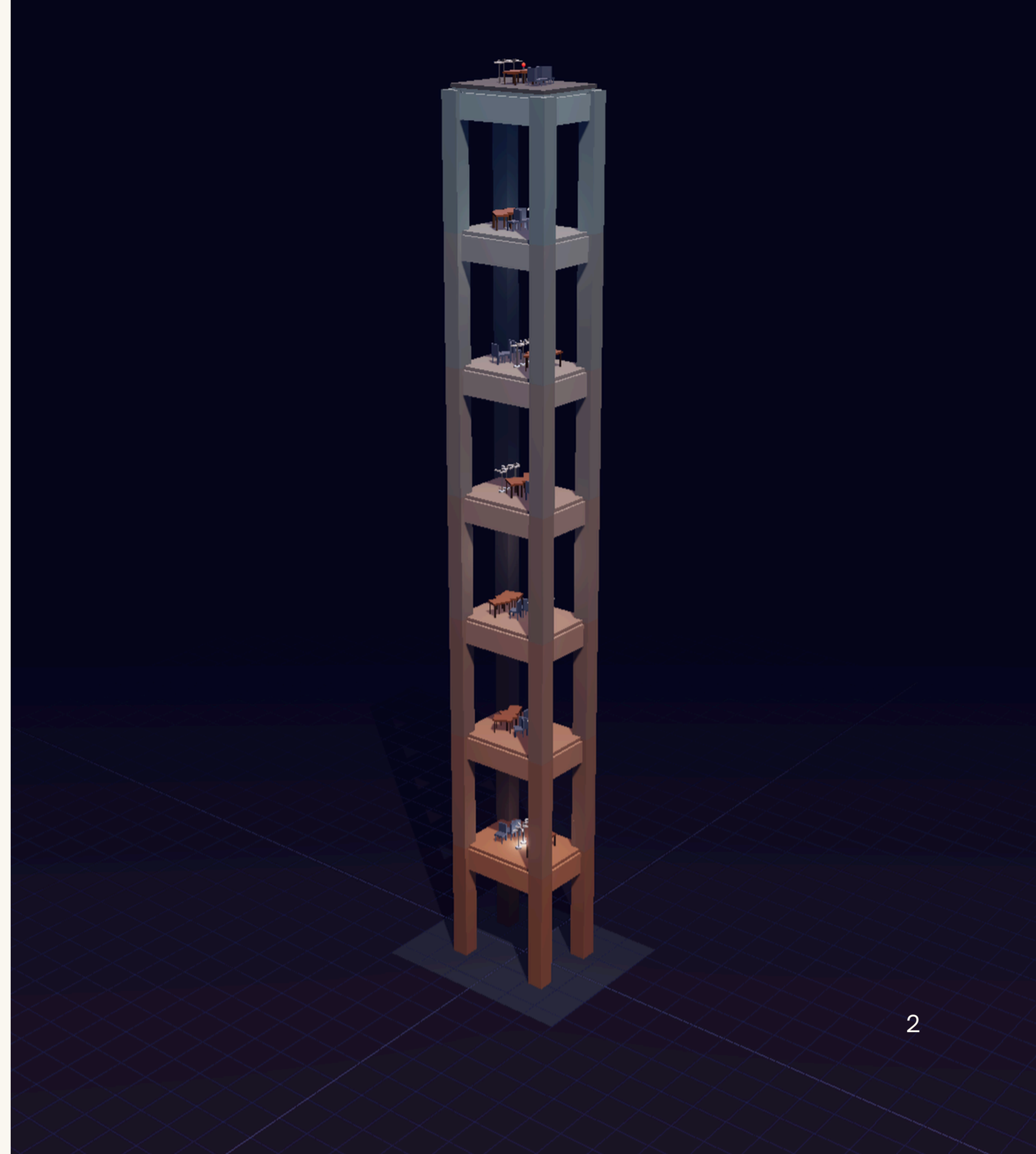
A 3D earthquake simulator. Pick a building and a real earthquake record, then watch each floor move the way the equations say it should.

What you can change

The building: stories, floor area, columns.

The earthquake: magnitude, epicenter.

The playback: speed, camera, slow motion.



Problem Identification and Motivation

Too complicated

Research tools

Structural dynamics tools like OpenSeesPy are hard to use unless you already have the engineering background.

Unrealistic

Easy simulators

Friendly tools drift away from the science. At best they rest on many assumptions. At worst they are only an animation.

Our goals

Simple

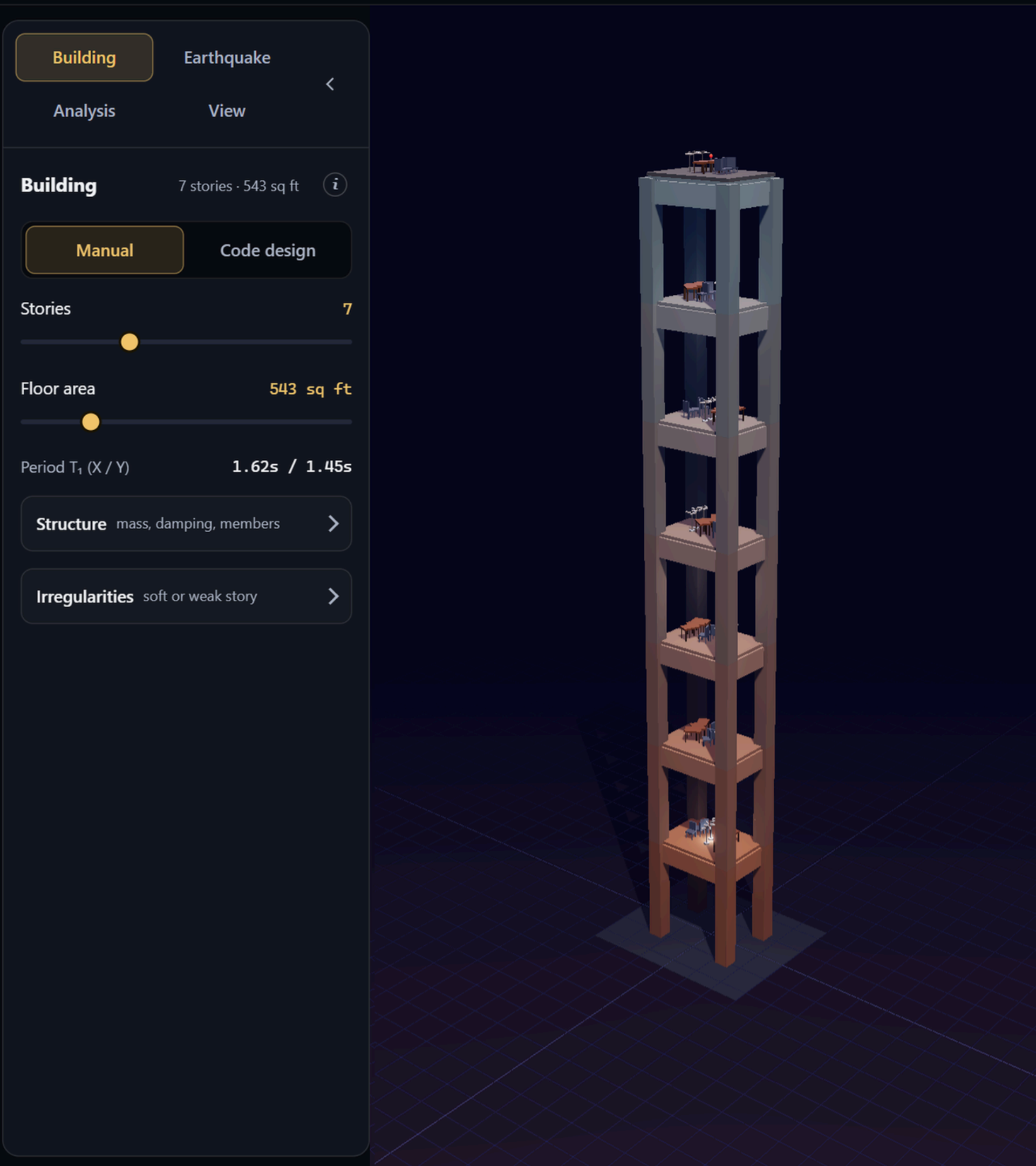
Anyone can use it. No background in buildings or structural dynamics needed.

Customisable

Many parameters to tune. Earthquake: magnitude and epicenter. Building: stories, floor area, column width.

Faithful

Keep the computation light, while using the algorithms engineers use in practice wherever we can.



FEATURES

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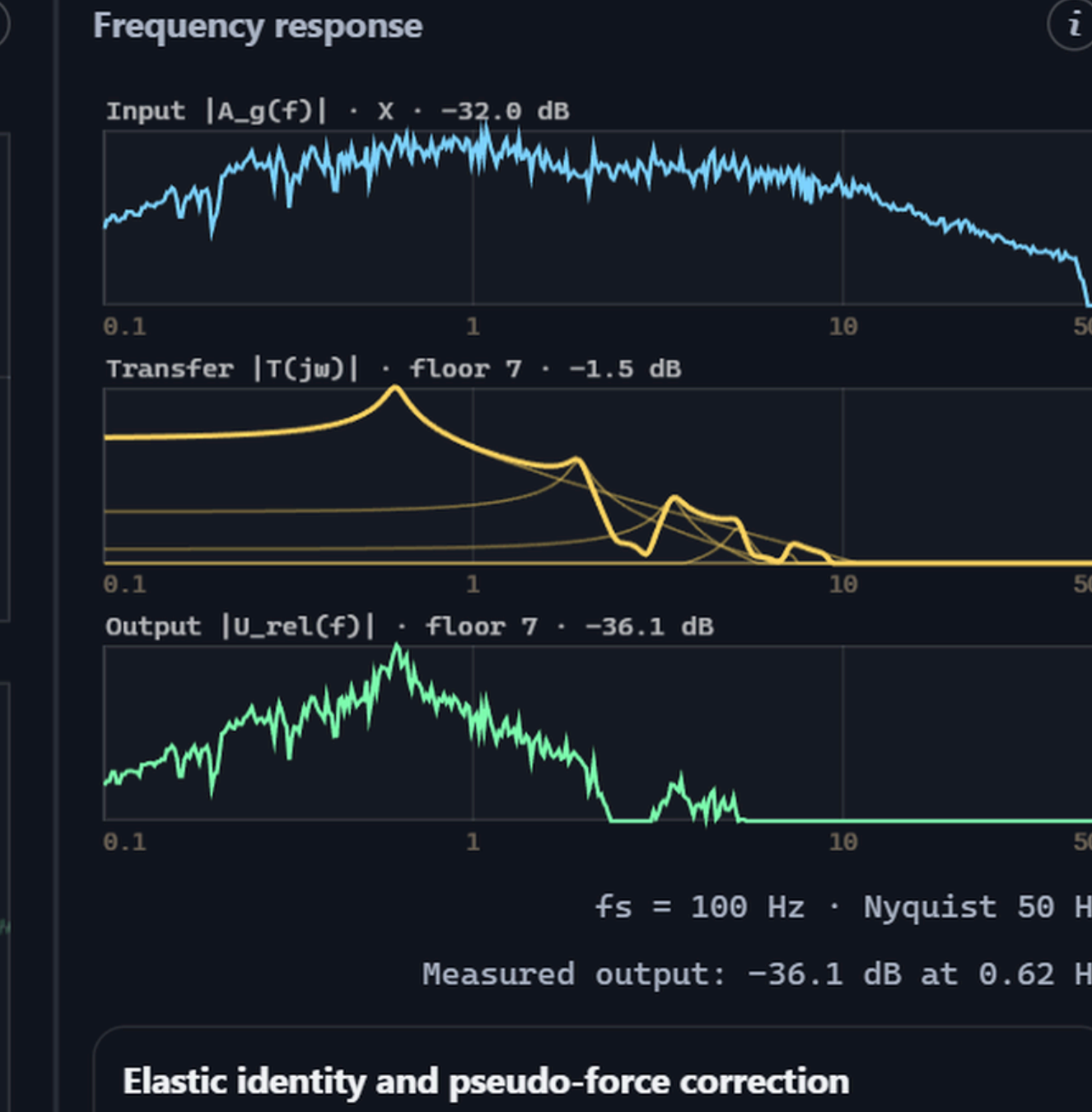
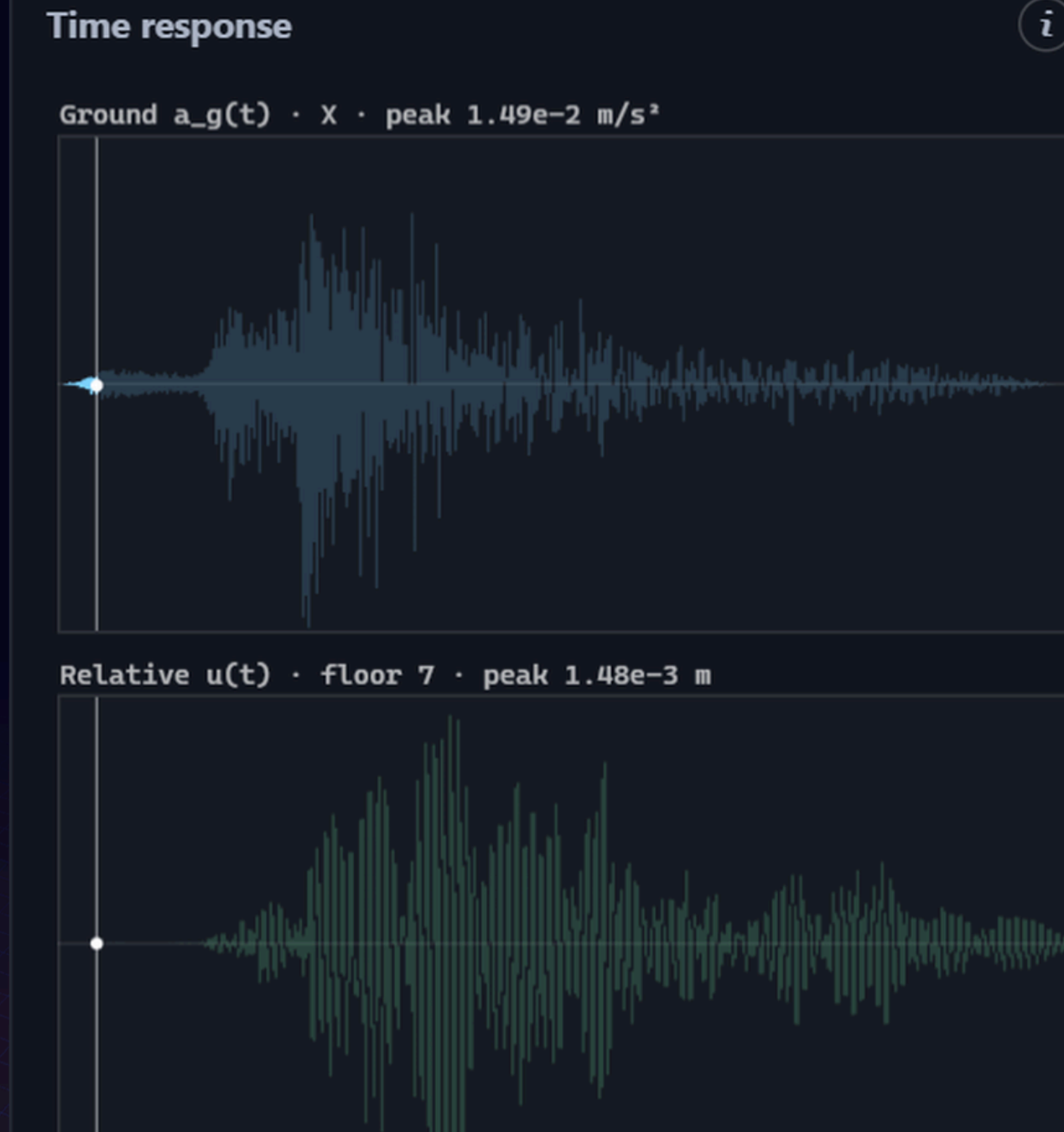
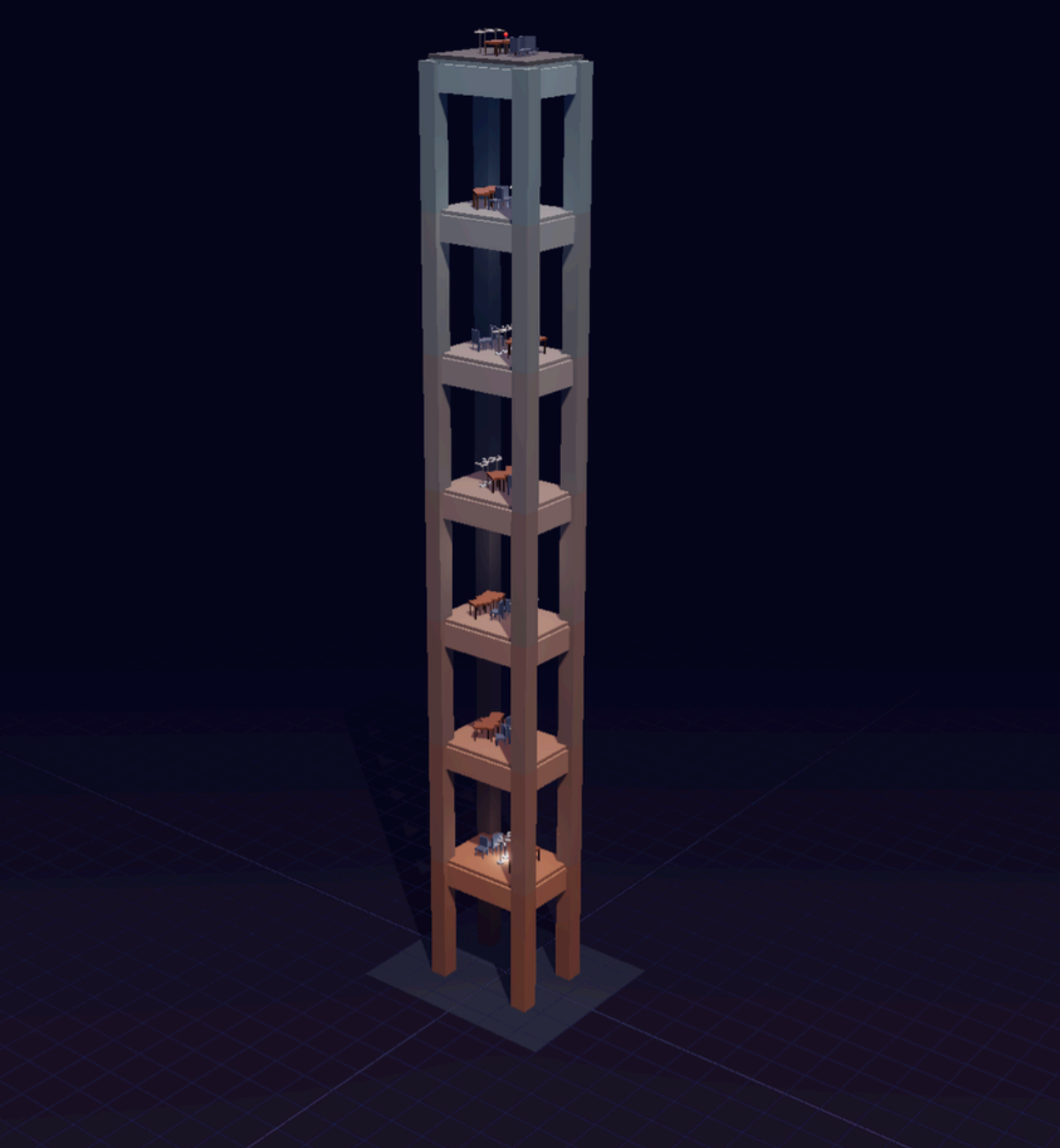
The controls

Building

Number of stories, floor area, mass, damping, and column and beam sizes. Code-design mode can size the members for you.

Earthquake

Ten real records from five earthquakes. Scale any of them by magnitude, epicenter distance and depth.



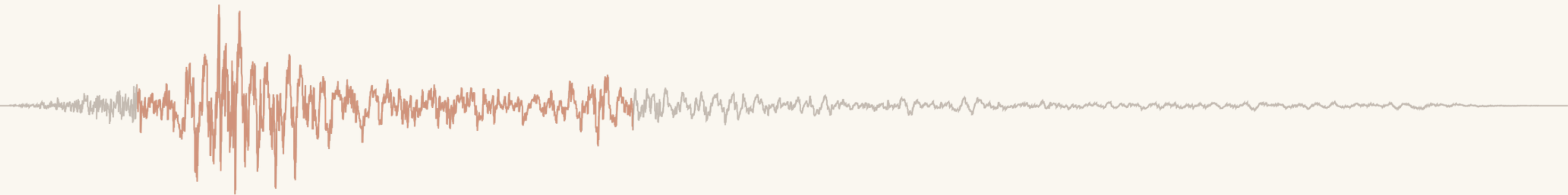
3D view: floors, columns and furniture move with the computed response.

Time: ground acceleration in, floor displacement out.

Frequency: input spectrum, transfer function, output spectrum.

NP.INTERP(1, -1)

Methodology



THE ELASTIC MODEL

Floors as masses, the ground as
the input, the Fourier transform as
the solver.

METHODOLOGY

Floors as spring-mass-dampers

One floor

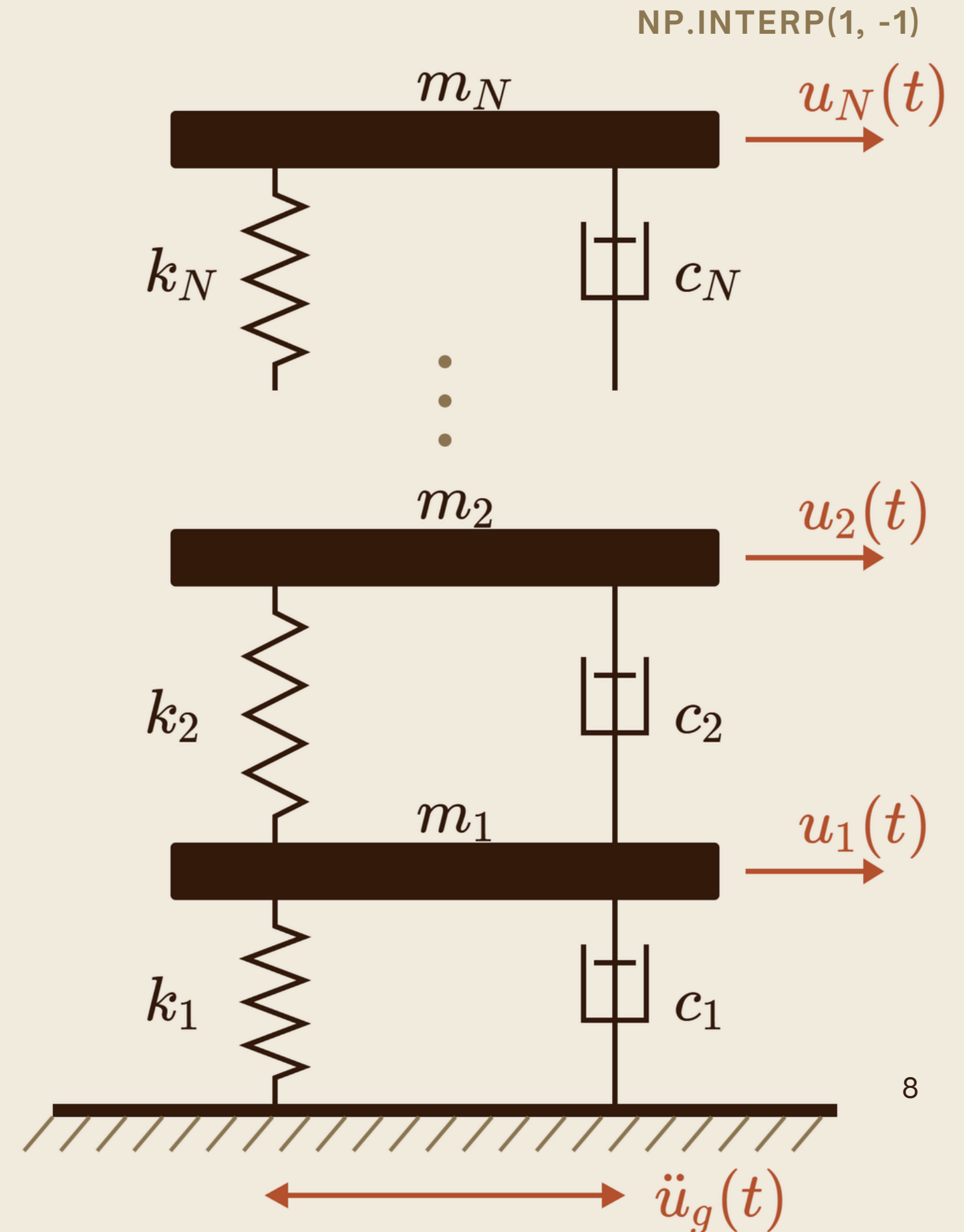
Each floor is a mass m . Its columns act as a spring k and a damper c . Newton's second law balances three internal forces against the applied load.

Equation of motion

$$f_I + f_D + f_S = f(t)$$

inertia damping stiffness

$$m\ddot{u} + c\dot{u} + ku = f(t)$$



The ground moves



Divide by m

No outside push

Nothing pushes the floor directly. Measure u relative to the ground, and the ground's acceleration appears as a force on the mass.

$$m\ddot{u} + c\dot{u} + ku = -m\ddot{u}_g(t)$$



the ground acts like a force on the mass

Two numbers per floor

The response now depends on the natural frequency ω_n and the damping ratio ζ .

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2u = -\ddot{u}_g(t)$$

$$\omega_n = \sqrt{k/m} \quad \zeta = \frac{c}{2m\omega_n}$$

Into the frequency domain

1 • Equation of motion, divided by m

$$m\ddot{u} + c\dot{u} + ku = -m\ddot{u}_g$$

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = -\ddot{u}_g$$

2 • Fourier transform: each derivative becomes a factor of $j\omega$

$$\dot{u} \rightarrow j\omega U(\omega) \quad \ddot{u} \rightarrow -\omega^2 U(\omega)$$

3 • Substitute into the equation

$$-\omega^2 U + j 2\zeta\omega_n\omega U + \omega_n^2 U = -A_g(\omega)$$

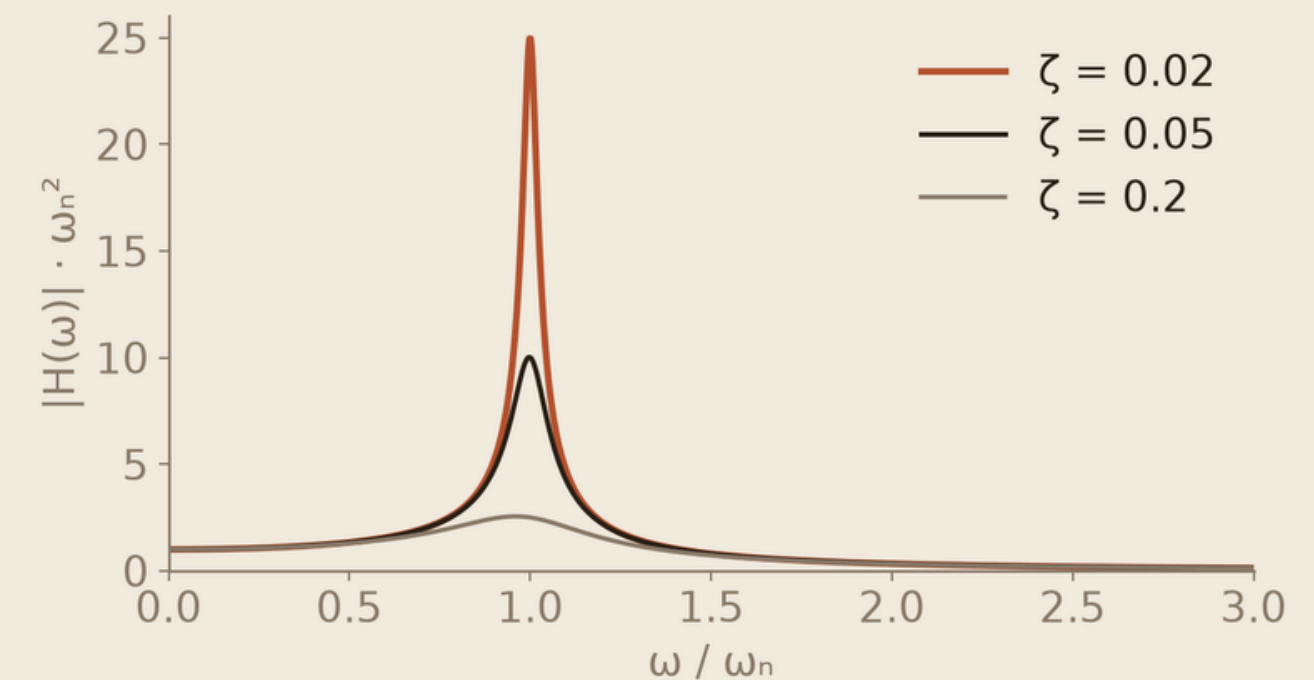
4 • Factor out $U(\omega)$

$$(\omega_n^2 - \omega^2 + j 2\zeta\omega_n\omega) U(\omega) = -A_g(\omega)$$

5 • Transfer function: output ÷ input, $H(\omega) = U(\omega) / A_g(\omega)$

$$H(\omega) = \frac{-1}{\omega_n^2 - \omega^2 + j 2\zeta\omega_n\omega}$$

It peaks near $\omega = \omega_n$. Less damping, taller peak (right).



Single-story frequency response $|H(\omega)|$

From an earthquake record to motion

01 Transform the earthquake record

$$A_g = \mathcal{F}\{\ddot{u}_g(t)\}$$

FFT the measured ground acceleration.

02 Apply the transfer function

$$U_{\text{rel}}(\omega) = H(\omega) A_g(\omega)$$

One multiplication at each frequency.

03 Return to time

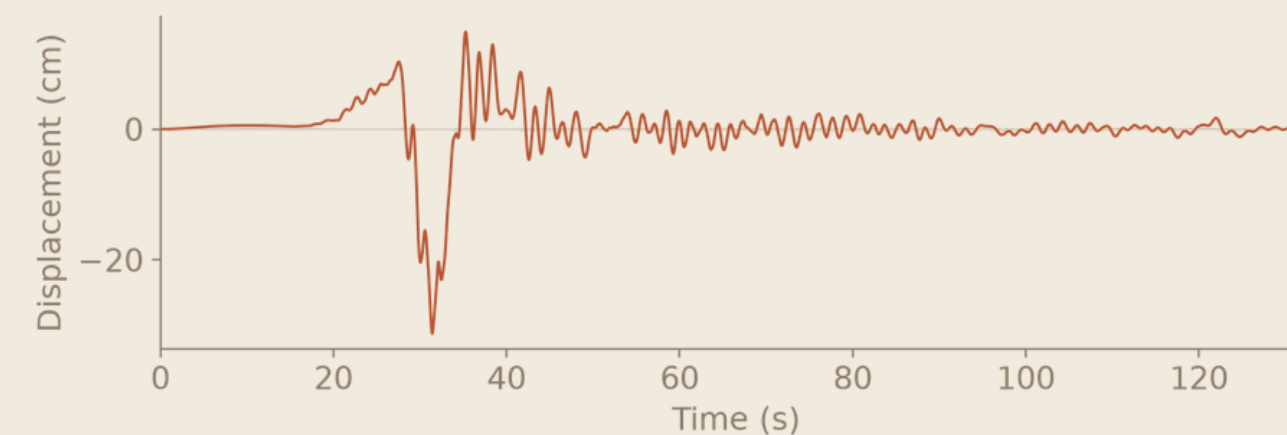
$$u_{\text{rel}}(t) = \mathcal{F}^{-1}\{U_{\text{rel}}(\omega)\}$$

The inverse FFT gives displacement at every sampled instant.

04 THE MOTION WE ANIMATE

$$u(t) = \underbrace{\mathcal{F}^{-1}\{H(\omega)A_g(\omega)\}}_{\text{relative response}} + \underbrace{u_g(t)}_{\text{ground motion}}$$

Computed roof displacement over time



The earthquake record becomes a full motion history.

Example output: 7-story model • 1999 Kocaeli, station ATK

METHODOLOGY

Domain Expansion

Multi-story: one mass becomes N masses, one equation becomes N coupled equations.



$N = 1$



$N = 20$

Coupled equations of motion

Newton's law, floor by floor

Floor 1

$$m_1 \ddot{u}_1 + c_1 \dot{u}_1 + k_1 u_1 - c_2(\dot{u}_2 - \dot{u}_1) - k_2(u_2 - u_1) = -m_1 \ddot{u}_g$$

⋮

Floor i

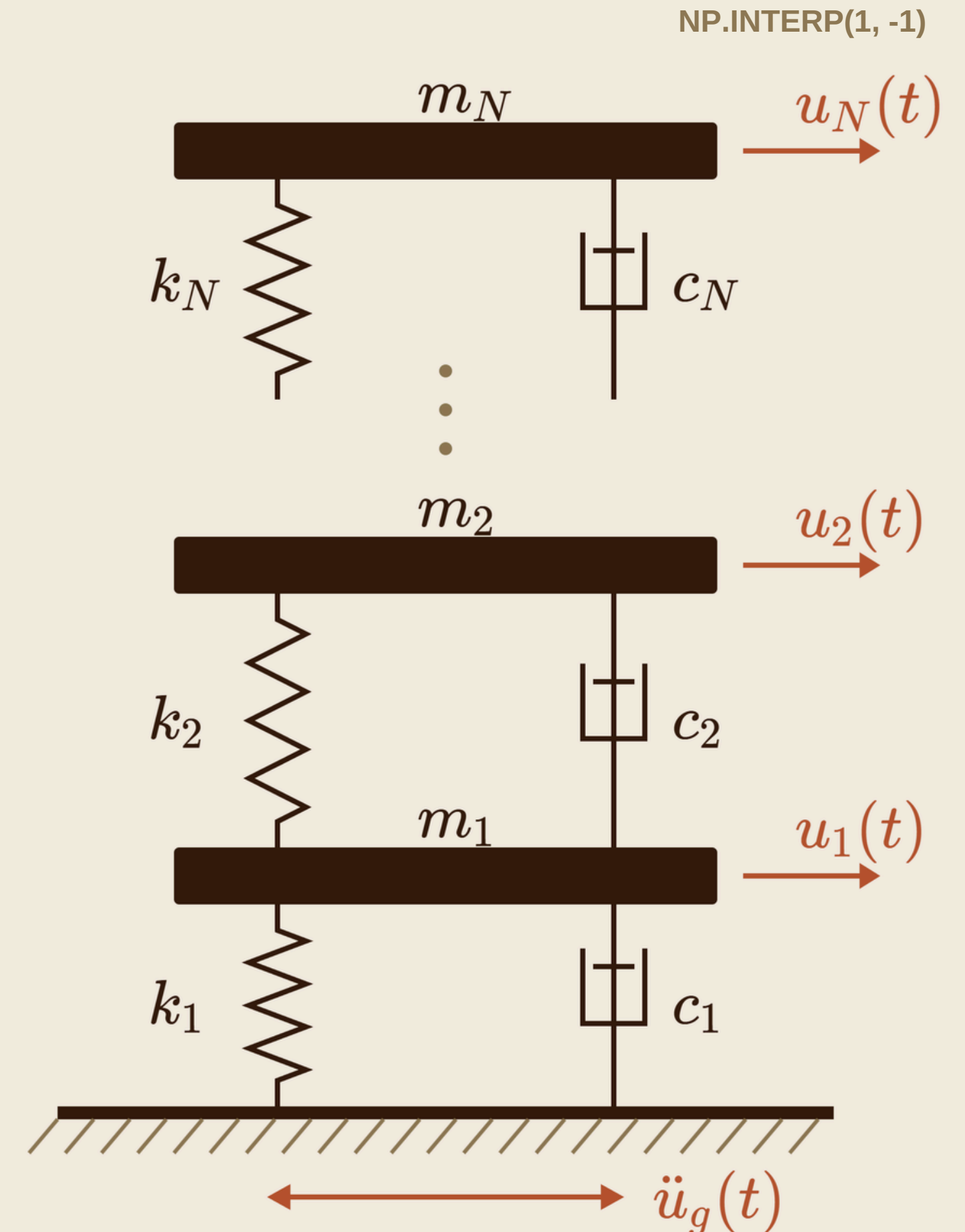
$$m_i \ddot{u}_i + c_i(\dot{u}_i - \dot{u}_{i-1}) + k_i(u_i - u_{i-1}) - c_{i+1}(\dot{u}_{i+1} - \dot{u}_i) - k_{i+1}(u_{i+1} - u_i) = -m_i \ddot{u}_g$$

⋮

Floor N

$$m_N \ddot{u}_N + c_N(\dot{u}_N - \dot{u}_{N-1}) + k_N(u_N - u_{N-1}) = -m_N \ddot{u}_g$$

The orange terms tie each floor to the floors above and below it, so no equation can be solved on its own.



Variables → matrices

Stack the floors into a vector

One unknown displacement per floor.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_N(t) \end{bmatrix}$$

Mass matrix

Each floor's mass sits on the diagonal. Floor i 's inertia depends only on its own acceleration, so every off-diagonal entry is zero.

$$\mathbf{M} = \begin{bmatrix} m_1 & 0 & \cdots & 0 \\ 0 & m_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_N \end{bmatrix}$$

Stiffness matrix

Why it is tridiagonal

A floor is joined only to the story below it and the story above it. So the spring force on floor i involves just three displacements, and row i of K has just three non-zero entries. Everything else in the row is zero.

$$\begin{aligned} f_{S,i} &= k_i (u_i - u_{i-1}) - k_{i+1} (u_{i+1} - u_i) \\ &= -k_i u_{i-1} + (k_i + k_{i+1}) u_i - k_{i+1} u_{i+1} \end{aligned}$$

$$\mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 & 0 & 0 \\ -k_2 & k_2 + k_3 & -k_3 & 0 \\ 0 & \ddots & \ddots & -k_N \\ 0 & 0 & -k_N & k_N \end{bmatrix}$$

The whole building in one line

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{1}\ddot{u}_g(t)$$

N coupled equations

One row per floor. The vector 1 applies the same ground acceleration to every floor.
The coupling sits in K.

Next step

Split this into N independent equations, one per mode. Eigenvectors do exactly that.

Eigenvalue solution

1

A coupled system: each y_i' depends on every y_j .

$$\mathbf{y}' = A\mathbf{y}$$

2

If A has n independent eigenvectors, it can be diagonalized.

$$A = PDP^{-1}$$

3

Change variables. Each equation now stands alone and solves as an exponential.

$$\mathbf{y} = P\mathbf{u} \Rightarrow \mathbf{u}' = D\mathbf{u}$$

$$u'_i = \lambda_i u_i \Rightarrow u_i = c_i e^{\lambda_i t}$$

Anton, §5.3

Move to the eigenvector basis and a coupled system becomes n separate equations. We do the same with M and K .

Why $\mathbf{K}\boldsymbol{\varphi} = \omega^2\mathbf{M}\boldsymbol{\varphi}$



What $\boldsymbol{\varphi}$ looks like

Guess a shape that oscillates

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\mathbf{u}(t) = \boldsymbol{\varphi} \sin \omega t \Rightarrow \ddot{\mathbf{u}} = -\omega^2 \boldsymbol{\varphi} \sin \omega t$$

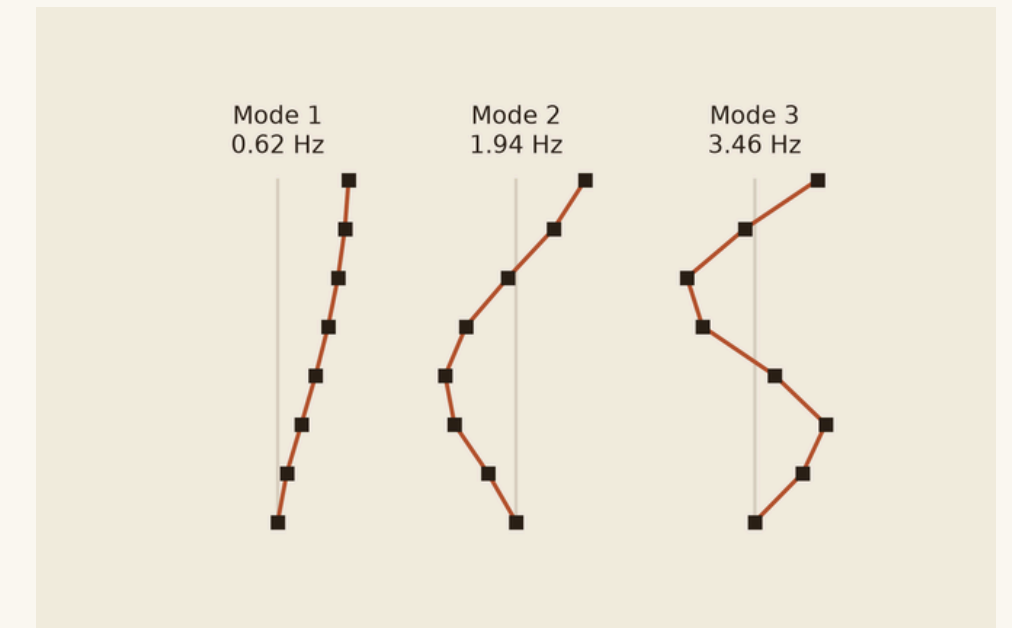
$$(\mathbf{K} - \omega^2\mathbf{M}) \boldsymbol{\varphi} \sin \omega t = \mathbf{0}$$

not zero for all t , so the bracket must vanish

$$\mathbf{K}\boldsymbol{\varphi} = \omega^2\mathbf{M}\boldsymbol{\varphi}$$

Mode shapes

Each eigenvector $\boldsymbol{\varphi}$ is a pattern the building sways in, at its own ω . These are the first three for our 7-story model.



Diagonalization

1



2



3

Stack the mode shapes into Φ
and write $\mathbf{u} = \Phi \mathbf{q}$.

Mass-normalise the modes. Φ
then turns $\mathbf{M}$ into $\mathbf{I}$ and $\mathbf{K}$ into a
diagonal matrix.

Each mode gets its own
equation, driven by Γ_i times the
ground motion.

$$\Phi = [\varphi_1 \quad \varphi_2 \quad \cdots \quad \varphi_N] \quad \mathbf{u} = \Phi \mathbf{q}$$

$$\Gamma_i = \varphi_i^T \mathbf{M} \mathbf{1}$$

$$\varphi_i^T \mathbf{M} \varphi_i = 1 \quad \Phi^T \mathbf{K} \Phi = \text{diag}(\omega_i^2)$$

$$\ddot{q}_i + 2\zeta_i \omega_i \dot{q}_i + \omega_i^2 q_i = -\Gamma_i \ddot{u}_g(t)$$

Result

N coupled equations become N single-story equations. Each one is the SDOF problem we already solved.

Per mode



Per floor

Same form as one story

Mode i responds like a single-story building with frequency ω_i and damping ζ_i .

$$H_i(\omega) = \frac{-\Gamma_i}{\omega_i^2 - \omega^2 + j 2\zeta_i\omega_i\omega}$$

↑
the single-story $H(\omega)$, scaled by Γ_i

Add the modes back

Weight each mode by its shape at floor r and sum. This is the transfer function the app plots for each floor.

$$U_r(\omega) = \sum_{i=1}^N \varphi_{ri} H_i(\omega) A_g(\omega)$$

Tying it together

1 · Equation of motion

N coupled equations, in time.

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{1} \ddot{u}_g(t) \longrightarrow H_i(\omega) = \frac{-\Gamma_i}{\omega_i^2 - \omega^2 + j 2\zeta_i \omega_i \omega}$$



2 · Transfer function

One per mode, from the eigen-solve.

3 · Multiply

Convolution in time is multiplication in frequency.

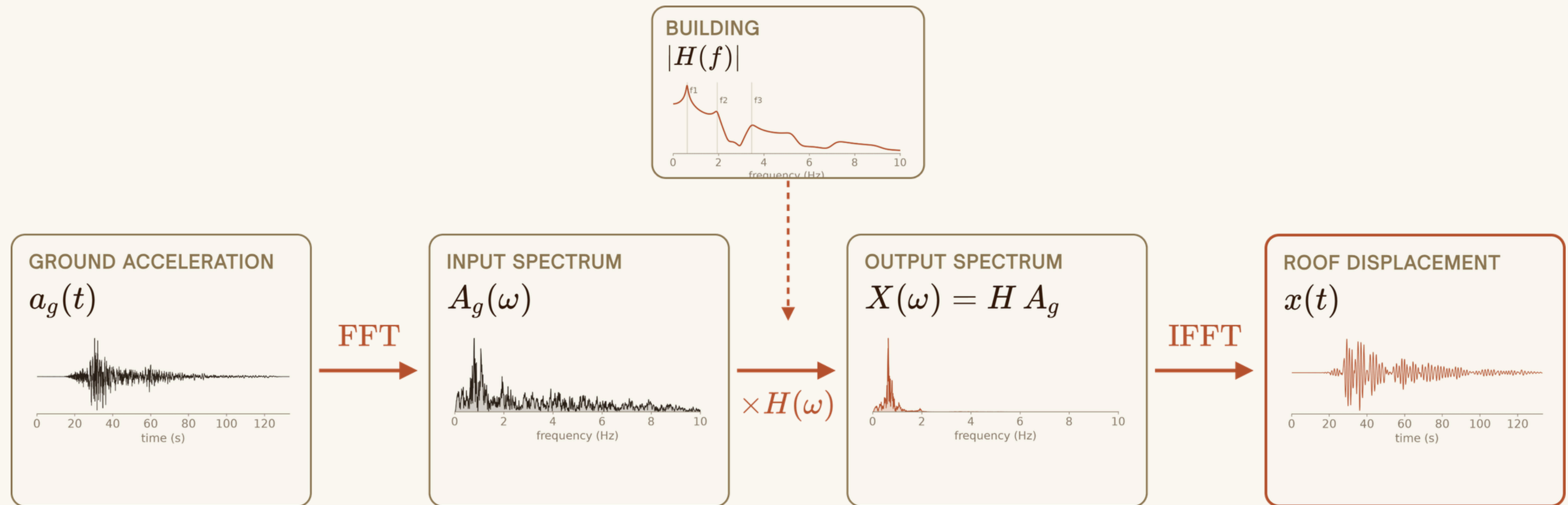
$$\longleftarrow X(\omega) = H(\omega) A(\omega)$$

4 · Back to time

The inverse FFT gives each floor's displacement history.

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

The pipeline



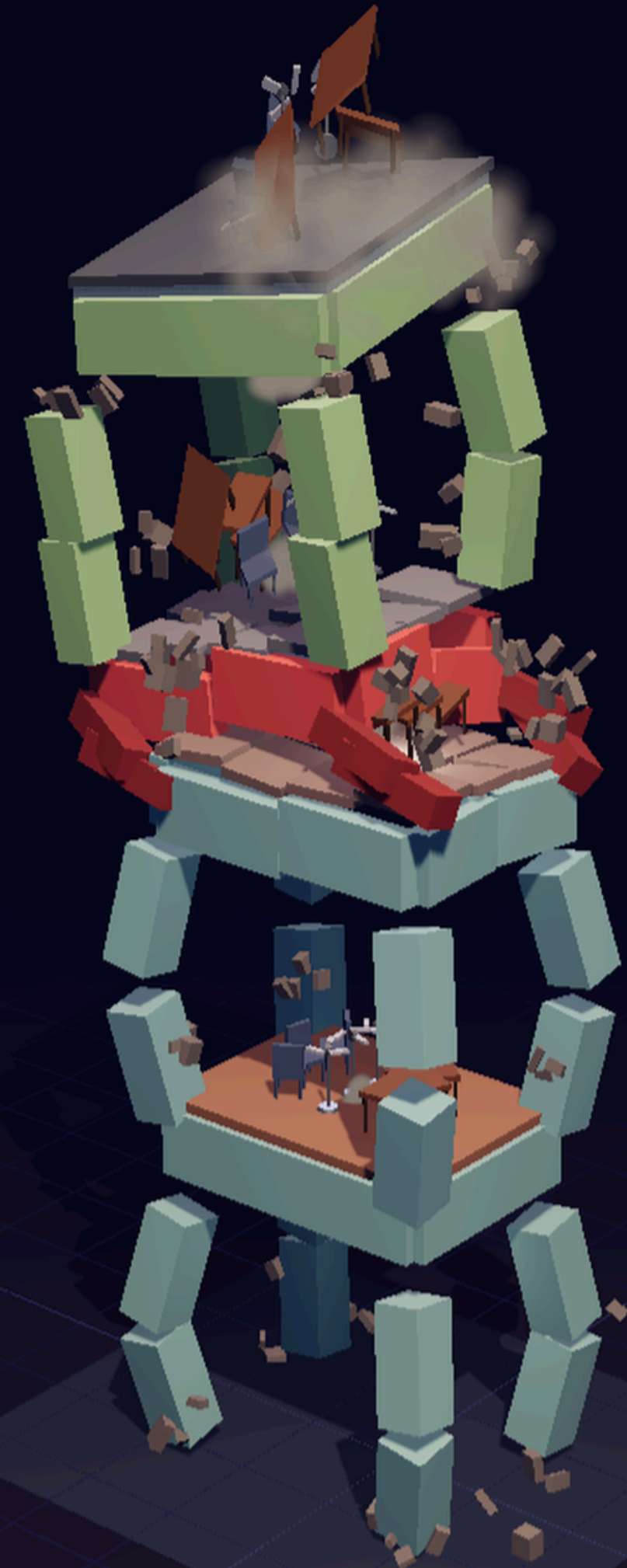
Real run: the 1999 Kocaeli record (station ATK) through our 7-story model. The app computes every floor's motion this way.

NP.INTERP(1, -1)

We've shaken it.

NOW, HOW DO WE BREAK IT?

Past the elastic limit, columns yield and the building's memory matters. Two slides, in brief.



Simulating collapse

Hysteresis

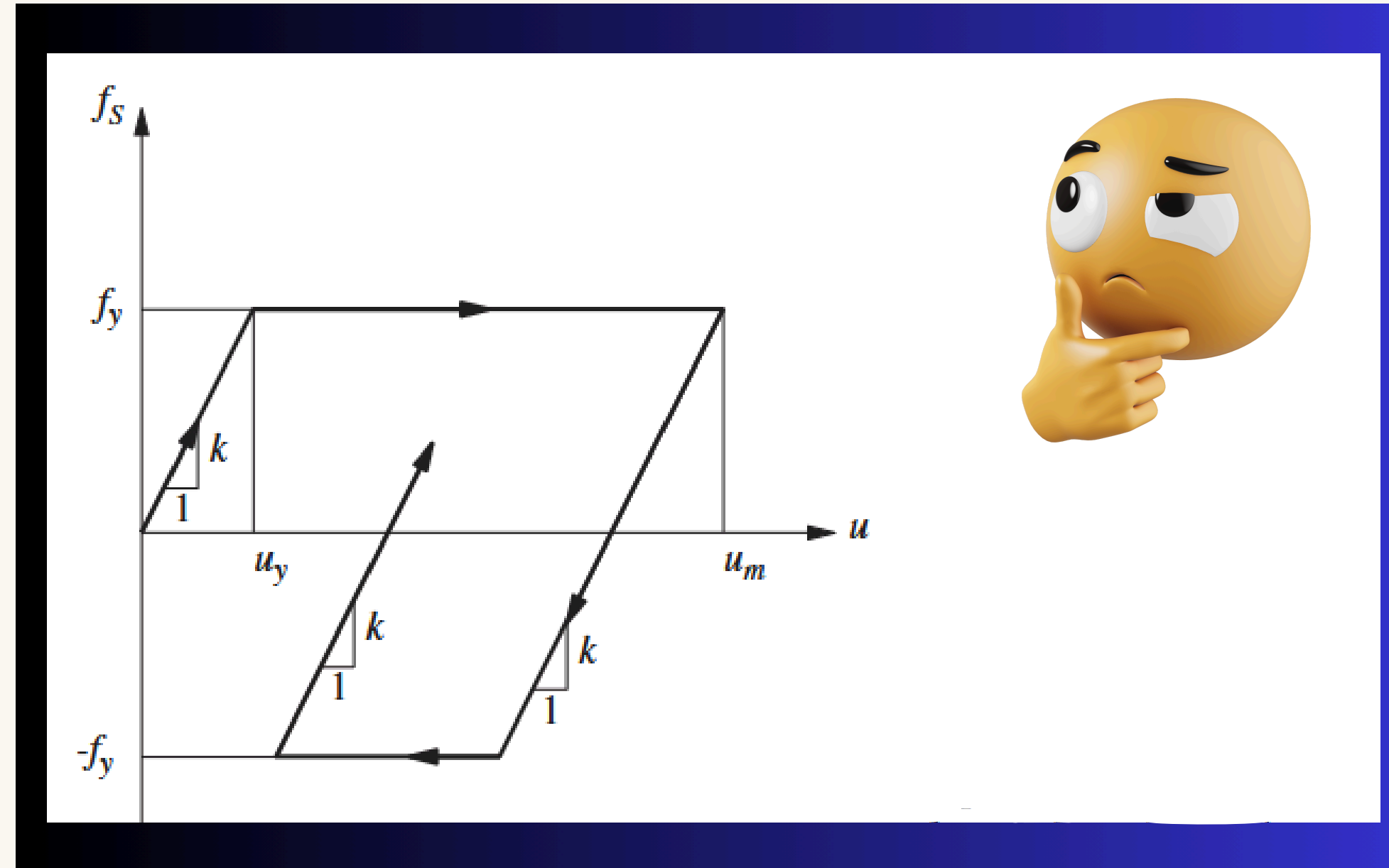
Past yield, a column's force depends on its whole drift history. Each loop leaves it weaker. Right: story 3 in our collapse demo, up to 80.5 s.

Keep the solver linear

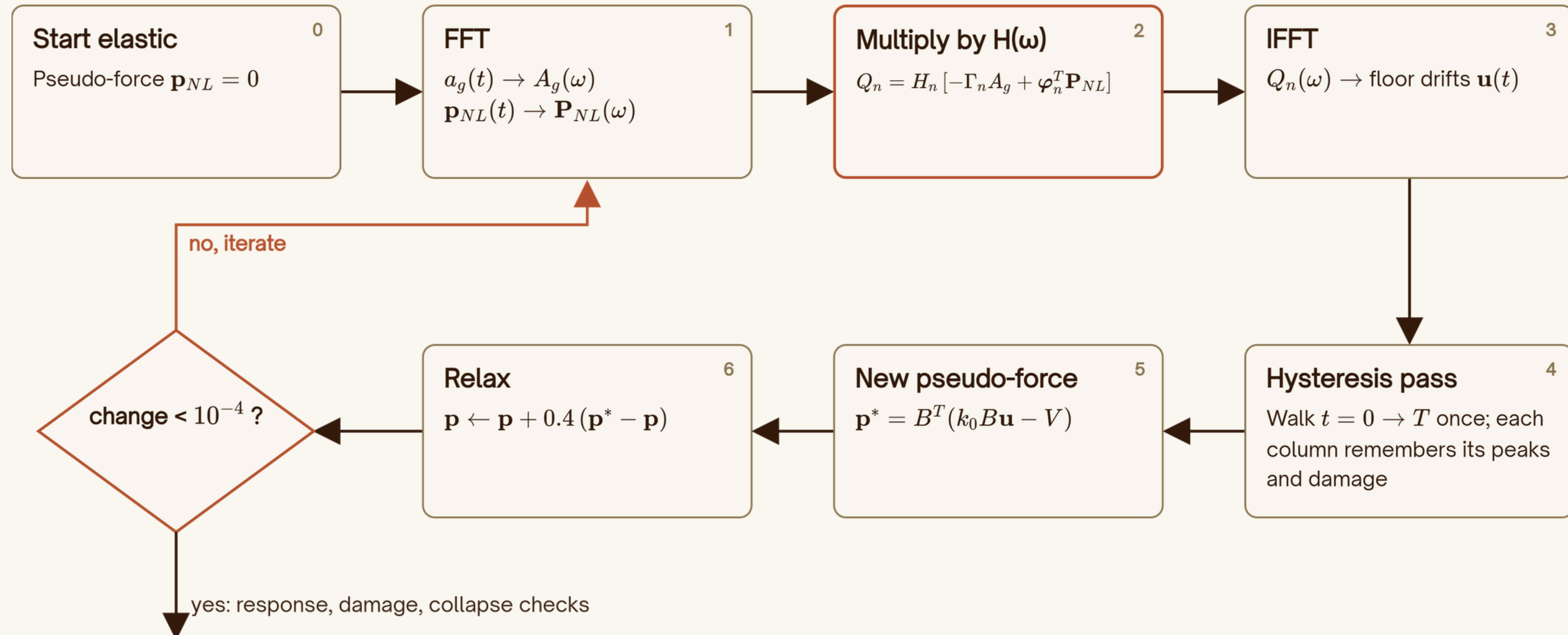
Move the nonlinear part to the right side as a pseudo-force. The left side stays linear, so the FFT solver still applies.

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}_L\mathbf{u} = -\mathbf{M}\mathbf{1} a_g + \mathbf{p}_{NL}$$

$$\mathbf{p}_{NL} = \mathbf{K}_L\mathbf{u} - \mathbf{F}_s(\text{history})$$



Hybrid frequency-time domain loop



History lives in step 4: one causal pass over the record per iteration. By default the whole record is one FFT block; overlap-save chunks are optional. The collapse demo converged in 3 iterations.

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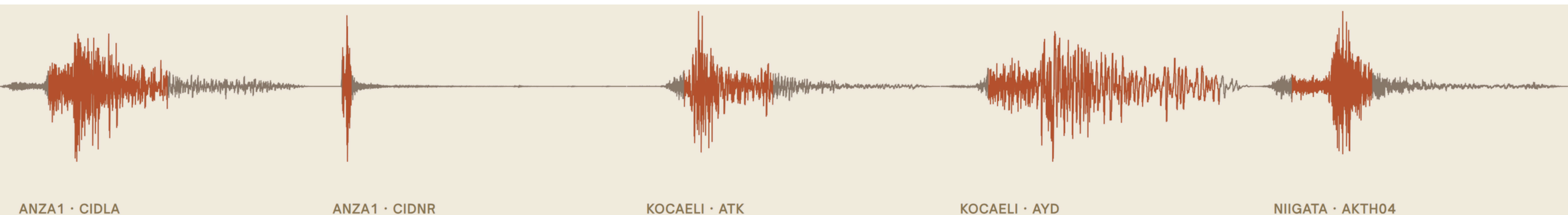
Acknowledgements

PEER

Pacific Earthquake Engineering Research Center, for the ground-motion records (.AT2 and .DT2) every simulation runs on.

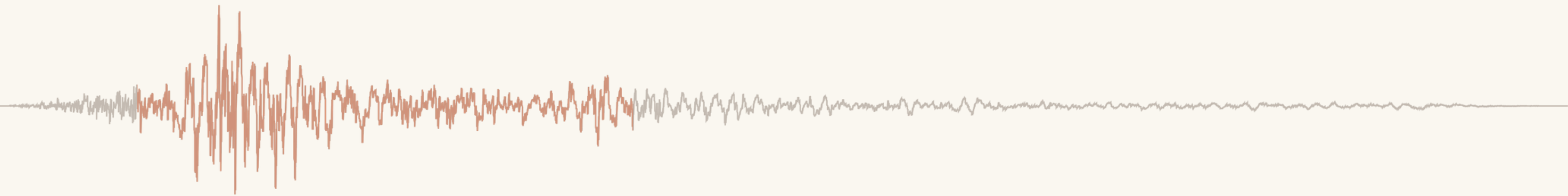
KiK-net

NIED's KiK-net strong-motion network, whose stations recorded the Niigata records we use.



LUCKY SEISMIC STRIKE

Thank you



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